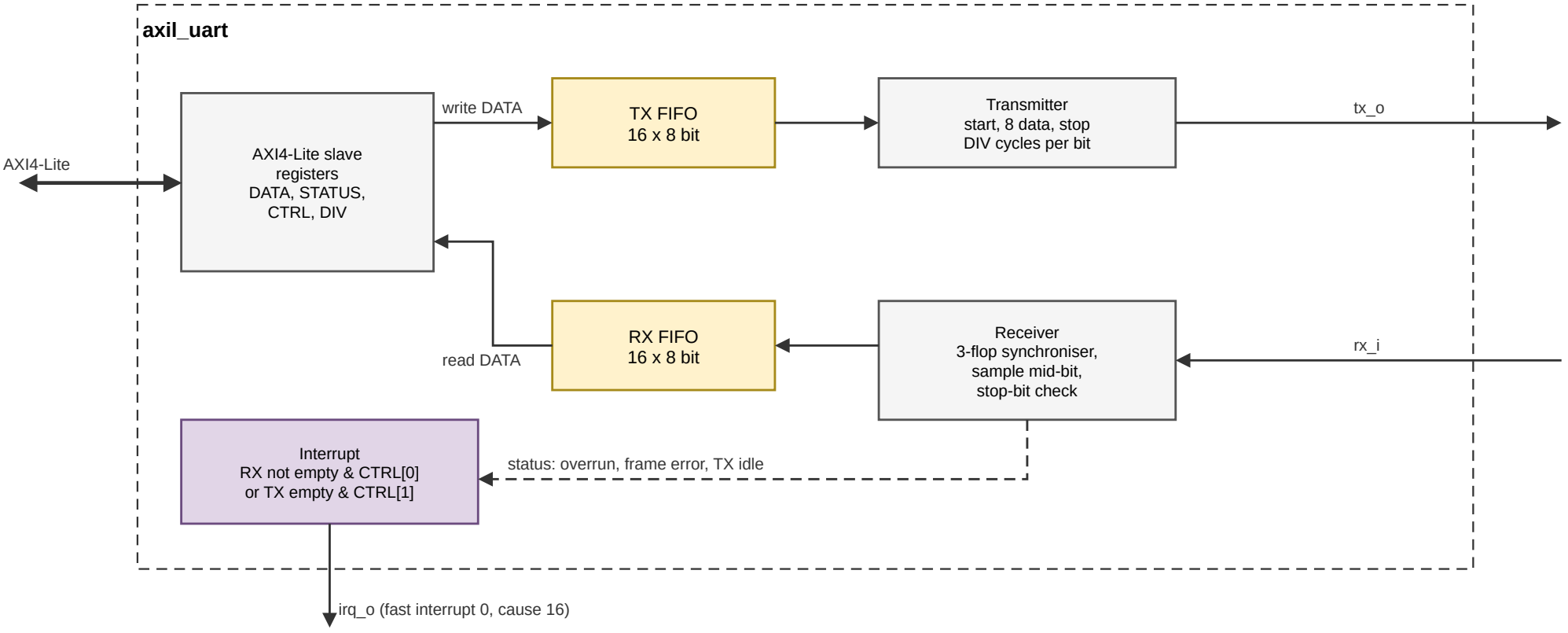
axil_uart
AXI4-Lite
AXI4-Lite slave
registers
DATA, STATUS,
CTRL, DIV
write DATA
TX FIFO
16 x 8 bit
Transmitter
start, 8 data, stop
DIV cycles per bit
tx_o
read DATA
RX FIFO
16 x 8 bit
Receiver
3-flop synchroniser,
sample mid-bit,
stop-bit check
rx_i
Interrupt
RX not empty & CTRL[0]
or TX empty & CTRL[1]
status: overrun, frame error, TX idle
irq_o (fast interrupt 0, cause 16)